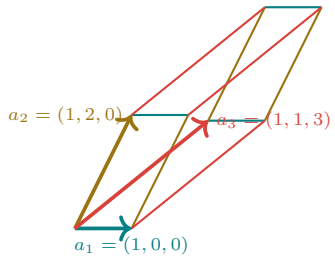
LINEAR ALGEBRA LADDER 5 OF 9

Determinants: area, volume and invertibility

The determinant measures how a matrix scales area or volume, and it vanishes exactly when the matrix flattens space.

1 / THE RULE AND THE PICTURE

The determinant of a 3×3 matrix is the volume of the box spanned by its columns



For the columns $a_1 = (1, 0, 0)$, $a_2 = (1, 2, 0)$, $a_3 = (1, 1, 3)$ the matrix is triangular with diagonal $1, 2, 3$, so the box has volume

$$\det \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{pmatrix} = 1 \cdot 2 \cdot 3 = 6$$

In two dimensions the same idea gives area, $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$, and the sign records whether orientation is kept or reversed.

2 / COMPUTING A 3×3 DETERMINANT / TWO WAYS

Expand along a row with signs $+, -, +$, or use the rule of Sarrus

For $A = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & 2 \\ 0 & 1 & 1 \end{pmatrix}$, expanding along the first row,

$$\det A = 2 \begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix} - 0 \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} + 1 \begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} = 2 \cdot 1 - 0 + 1 \cdot 1 = 3.$$

The rule of Sarrus adds the three down-right diagonals and subtracts the three down-left ones: $(2 \cdot 3 \cdot 1 + 0 \cdot 2 \cdot 0 + 1 \cdot 1 \cdot 1) - (1 \cdot 3 \cdot 0 + 2 \cdot 2 \cdot 1 + 0 \cdot 1 \cdot 1) = 7 - 4 = 3$.

3 / THE RULES THAT MAKE IT WORK

Row operations change the determinant in a predictable way

operation on $A = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & 2 \\ 0 & 1 & 1 \end{pmatrix}$	effect on $\det$	example
swap two rows	sign changes	-3
multiply a row by k	determinant times k	6 for $k = 2$
add a multiple of one row to another	unchanged	3
two rows proportional	0	the $1 \ldots 9$ matrix gives 0
product	$\det(AB) = \det A \det B$	$\det A^2 = 9 = 3^2$
triangular matrix	product of the diagonal	$1 \cdot 2 \cdot 3 = 6$

These rules explain why elimination computes determinants: reduce to a triangular matrix, track the sign of each swap and the factor of each scaling, and multiply the diagonal.

4 / CRAMER'S RULE / RATIOS OF AREAS

Each unknown is a ratio of two determinants

For $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$ we have $\det A = 5$, and replacing a column by b gives

$$x = \frac{\begin{vmatrix} 5 & 1 \\ 5 & 3 \end{vmatrix}}{5} = \frac{10}{5} = 2, \qquad y = \frac{\begin{vmatrix} 2 & 5 \\ 1 & 5 \end{vmatrix}}{5} = \frac{5}{5} = 1,$$

because the parallelogram spanned by b and one column is x or y times the parallelogram of the two columns, and the formula needs $\det A \neq 0$.

5 / ZERO DETERMINANT / FLATTENING

$\det A = 0$ exactly when the columns are dependent, and then A has no inverse

The matrix $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ has $\det = 0$ because the third row is $2 \cdot (4, 5, 6) - (1, 2, 3)$, so the box collapses to a plane and every image lies in it. Then $Ax = b$ has no solution or infinitely many, and an invertible matrix is exactly one whose determinant is non-zero.

ASIDE / SHEARS KEEP VOLUME

Adding a multiple of one row to another slides the box sideways without changing its volume, which is the rule used in elimination.

LINKS TO OTHER SHEETS

The guises of the determinant are in *The determinant in its many guises*; the previous sheet is *Inverses and elementary matrices* and the next is *Span, basis and the four subspaces*.

Convention. $\det A$ is the signed volume scale; $|\ |$ around a matrix means its determinant.

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